RESEARCH ARTICLE

Agri-Vision: AI Powered Crop Advisory Platform

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Abstract

Agriculture is witnessing a transformative shift with the integration of artificial intelligence, enabling data-driven solutions to long-standing challenges. Our project, *Agri Vision*, introduces an AI-powered crop advisory platform designed to assist farmers in improving crop productivity, resource utilization, and decision-making. By employing a diverse set of machine learning and deep learning models, the system offers personalized recommendations across multiple domains including crop and fertilizer suggestion, yield and rainfall prediction, as well as real-time pest and plant leaf disease detection through image classification.

The proposed system incorporates models such as Random Forest, Decision Tree, and Linear Regression for tabular predictions, achieving accuracies ranging from 90% to 95%, while convolutional neural networks (CNNs) deliver up a good accuracy in image-based tasks. Additionally, an AI-driven chatbot enhances user interaction by addressing farming-related FAQs, improving accessibility. This multi-module framework signifies a practical and scalable approach to precision agriculture, demonstrating high accuracy and adaptability, and laying the groundwork for intelligent, sustainable farming practices.

Keywords: Crop Recommendation, Fertilizer Prediction, Convolution Neural Network (CNN), AI Chatbot, Yield Prediction, Deep Learning.

INTRODUCTION

1.1 Introduction to Problem Domain

Agriculture is a cornerstone of food security and economic growth. However, farmers often struggle to make timely decisions due to a lack of real-time data and predictive insights. *Agri Vision* is an AI-powered platform designed to address these challenges.

- Personalized Recommendations based on real-time data and advanced analytics are crucial for optimizing productivity.
- Efficient Resource Allocation, through resource optimization, is essential for reducing waste and promoting sustainable farming.
- Enhanced Productivity using AI-driven tools enables farmers to make faster, smarter decisions, leading to increased crop yields

1.2 Aim of the Project

To develop a robust AI-powered system that leverages Machine Learning and Deep Learning models to provide accurate and personalized agricultural recommendations. The system is designed to accept various inputs and deliver insights such as suitable crop selection, fertilizer recommendation, yield and rainfall prediction, and disease detection, thereby supporting farmers in making data-driven decisions to enhance productivity and sustainability in farming.

1.3 Objectives of the Project

Our objective is to comprehensively address key challenges in modern agriculture by harnessing the power of Artificial Intelligence and Machine Learning. The project is structured around the following core objectives:

Optimize Crop Yield: The primary objective is to implement predictive models that estimate crop yields based on inputs such as weather data, soil properties, and crop type. This enables farmers to make timely and informed decisions regarding planting and harvesting, ultimately maximizing agricultural output.

Improve Crop Selection: Focuses on recommending the most suitable crops based on soil composition, climate conditions, and nutrient availability. By aligning crop choices with environmental factors, the system helps reduce the risk of crop failure and increases productivity and profitability.

Enhance Fertilizer Usage: Developing a recommendation system to guide farmers in optimal fertilizer usage. The objective is to prevent both underuse and overuse by analyzing soil nutrient levels and crop requirements, ensuring efficient resource use and sustainable agricultural practices.

Detect Plant Leaf Diseases Early: Utilize deep learning- based image classification models to identify plant leaf diseases at an early stage from leaf images. Timely detection and suggested treatments aim to minimize crop damage and prevent the spread of infections.

Predict Weather Conditions: Offering weather updates based on the city name, informing the user about temperature, humidity, and rainfall predictions, aiding in farming decisions and scheduling.

Develop Farmer Chatbot: Designing an AI-driven chatbot to provide quick and personalized responses to farmer's queries regarding crop care, weather, government schemes, and market trends. The chatbot enhances accessibility to agricultural knowledge and decision-making support.

1.4 Applications

The AI-powered Agri Vision platform holds immense potential across various agricultural domains. In crop yield optimization, it empowers farmers to make informed decisions about sowing and harvesting by offering accurate yield predictions based on environmental and soil factors. This proactive approach helps in minimizing losses and maximizing productivity.

For crop selection, the system supports sustainable farming practices by recommending crops best suited to specific soil types and climatic conditions, thereby reducing the risk of crop failure and promoting better land utilization and resource efficiency. In terms of fertilizer management, the project aids farmers in optimal fertilizer application by analyzing soil nutrient levels and crop demands, thus preventing overuse that can harm the environment and underuse that may reduce yield, promoting environmentally conscious and cost-effective farming. The plant leaf disease detection feature plays a vital role in early-stage disease management by leveraging deep learning techniques to analyze leaf images, ensuring timely diagnosis and suggesting preventive or corrective measures to minimize crop loss and prevent widespread infections. With weather prediction capabilities, the platform enhances agricultural planning by providing localized forecasts for rainfall, temperature, and humidity, assisting farmers in scheduling irrigation, crop protection, and choosing ideal sowing or harvesting days. Additionally, the integration of a farmer-friendly AI chatbot boosts real- time support and accessibility by offering personalized

guidance on crop care, seasonal planning, government schemes, and market insights, thus reducing dependency on intermediaries and empowering farmers with essential knowledge.

Overall, this project promotes smart farming practices by combining advanced technology with user-centric design, paving the way for precision agriculture, sustainable resource usage, and enhanced decision-making in the agricultural sector.

1.5 Existing Solution methods

AI-powered agricultural advisory systems have increasingly adopted Machine Learning and Deep Learning techniques to improve decision-making in farming practices. These systems commonly utilize algorithms such as Decision Trees, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest, and ensemble learning for tasks like crop yield prediction, crop selection, and fertilizer recommendation. Deep learning models, particularly Convolutional Neural Networks (CNNs), are widely used for image-based plant leaf disease detection due to their high accuracy in visual classification. Furthermore, time-series models and regression-based techniques are employed for forecasting weather and rainfall patterns. These models enable the automation of agricultural insights by analyzing environmental data, soil characteristics, and crop histories, ultimately aiding farmers in optimizing output, reducing resource waste, and making informed decisions.

Despite their effectiveness, these systems face several limitations. The performance of ML/DL models in agriculture heavily relies on the availability and quality of datasets. Additionally, many models lack adaptability to changing climate conditions and localized farming practices. Algorithmic challenges such as overfitting in decision trees, high computational cost in KNN, and sensitivity to noise in SVM further affect model robustness. Moreover, deep learning models often require significant computational resources and expertise, making them difficult to deploy in low-resource settings. Accessibility remains a concern, as farmers in rural or underdeveloped areas may lack internet access, digital literacy, or the devices needed to utilize these platforms effectively. These constraints emphasize the need for more generalized, scalable, and farmer-friendly agricultural advisory solutions.

1.6 Proposed Solution Methods

The proposed Agri Vision: AI-Powered Crop Advisory Platform is a holistic, AI-driven system designed to revolutionize modern agriculture by combining intelligent automation with user-centric design. At the core of this platform lies an innovative farmer-centric approach—an AI-powered chatbot—uniquely developed to provide personalized, real-time assistance to farmers. Unlike traditional advisory systems that often lack accessibility and interactivity, this intelligent chatbot bridges the digital divide by offering instant responses to queries related to crop selection, fertilizer use, disease management, and weather forecasting, thus transforming the way farmers access expert guidance.

A standout feature of the system is its seamless website integration, which brings together all functionalities—including crop yield prediction, fertilizer recommendation, disease detection, weather forecasting, and expert advisory—under a single digital umbrella. This unified platform eliminates the need for farmers to navigate multiple tools or resources. With just one click, users can access comprehensive support, making the solution not only highly functional but also incredibly convenient and accessible, especially in rural regions where digital literacy may be limited.

From a technical standpoint, the platform incorporates a multi-model architecture built on advanced preprocessing strategies and hybrid learning methods. Structured data—like soil properties, temperature, and crop attributes—is pre-processed using normalization and entropy-based feature extraction to ensure cleaner, more informative inputs. Ensemble machine learning models such as Random Forest, Decision Trees, and K-Nearest Neighbors are employed for tasks like yield prediction and fertilizer selection, balancing both accuracy and interpretability. For visual tasks such as plant leaf disease detection, deep learning techniques like Convolutional Neural Networks (CNNs) are leveraged due to their ability to capture fine-grained image features. Use of real-time APIs enhanced the system's ability to provide localized weather and rainfall predictions.

Altogether, the Agri Vision platform delivers a powerful combination of precision, accessibility, and intelligent support. By integrating advanced AI models with an intuitive, interactive chatbot and a centralized web interface, the platform offers a one-stop digital solution for modern farming—empowering farmers to make smarter decisions, improve productivity, and embrace sustainable practices with ease.

1.7 Limitations

While the proposed Agri Vision platform presents a comprehensive and intelligent approach to agricultural advisory, certain challenges remain that offer scope for future enhancement. Its performance can be influenced by the availability of diverse, region-specific datasets, which are essential for accurate and localized predictions. Adapting to the dynamic nature of agricultural regulations across different regions is another key consideration, requiring regular updates to maintain relevance. Moreover, limited digital infrastructure and technical access in rural areas may impact adoption, though the platform’s user-friendly design and multilingual support aim to address this. These aspects highlight opportunities for scaling the system to better serve a wider farming community.

These aspects do not hinder the current effectiveness of the platform but rather highlight avenues for further scalability and improvement. With continued development,

integration of larger datasets, and collaboration with local agricultural bodies, Agri Vision: AI Powered Crop Advisory Platform has the potential to become an even more inclusive and adaptive tool for smart farming.

1.8 Scope

1.8.1 Context

The aim of the project is to develop an intelligent, AI-powered agricultural advisory platform that provides personalized and data-driven recommendations to farmers. By leveraging a combination of Machine Learning and Deep Learning models, the system supports crucial tasks such as crop yield prediction, crop and fertilizer recommendation, plant leaf disease detection, and weather forecasting. The platform is designed to be robust, scalable, and adaptable to various agricultural scenarios while ensuring ease of use through an intuitive interface and chatbot support.

1.8.2 Information Objectives

The software primarily addresses the following use cases:

a. Smart Agricultural Advisory: The platform accurately predicts crop yield, recommends suitable crops and fertilizers, detects plant leaf diseases and pests, and forecasts rainfall based on soil, weather, and crop data.

b. User-Friendly Accessibility: Ensures ease of use for farmers through a simple interface, regional language support, and interactive chatbot, promoting adoption across varying levels of digital literacy.

2. LITERATURE SURVEY

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have significantly influenced the field of smart agriculture by enabling data-driven solutions for complex farming challenges. The literature reviewed in this study reflects the growing focus on leveraging these technologies for enhancing productivity, supporting sustainable practices, and empowering farmers with actionable insights. This survey is structured into two main areas: **data preprocessing** and **model development**.

The insights derived from this literature form the foundation for our project, Agri Vision: AI-Powered Crop Advisory Platform, which aims to integrate multiple functionalities into a unified, farmer-friendly decision-support system.

Crop Selection and Yield Prediction using Machine Learning Approach – 2023

The paper explores how advancements in machine learning and smart computing can enhance decision-making in agriculture, particularly in Crop Yield Prediction (CYP) and crop classification.

The authors emphasize that traditional methods such as historical averages fail to incorporate multiple environmental parameters, often resulting in inaccurate predictions. To overcome this, the study implements machine learning techniques using environmental features like soil content, humidity, rainfall, and cultivation area to improve yield forecasting accuracy. The research involves the development and analysis of both regression and classification models for yield estimation and crop recommendation, respectively.

For yield prediction, Random Forest Regression was identified as the most effective model, achieving a Mean Absolute Error (MAE) of 0.64 and an R^2 score of 0.96, indicating a strong fit. In the case of crop prediction, Naïve Bayes classifier achieved an impressive accuracy of 99.39%, outperforming other classification algorithms tested in the study.

The dataset used was specific to districts in Maharashtra, and the authors highlight that the inclusion of real-time IoT-based data, irrigation and fertilizer inputs, and paid datasets could further improve the model's accuracy and reliability.[1]

Crop Prediction using Machine Learning Approaches – 2020

The study focuses on developing a crop recommendation system that leverages machine learning algorithms to assist farmers in choosing the most suitable crop for cultivation based on land and weather parameters. The motivation behind this work stems from the observation that many Indian farmers tend to cultivate the same crop repeatedly and apply fertilizers without proper knowledge of nutrient deficiencies, which adversely impacts both crop yield and soil health.

The system proposed in this study uses environmental features such as soil pH, temperature, humidity, and annual rainfall to recommend appropriate crops. Notably, Support Vector Machine (SVM) is used to predict annual rainfall based on historical data, while the remaining parameters are provided by the user. Based on these inputs, the system recommends the most suitable crop along with estimated seed quantity per acre, market price, and expected yield. Additionally, it provides recommendations based on the input nutrient values and the requirements of the suggested crop.

The experimental results show that the system performs effectively across different soil and environmental conditions, and the predictions are made accessible through a Graphical User Interface (GUI), making the system farmer friendly. The study highlights the importance of incorporating technological solutions to enhance decision-making in agriculture and reduce economic losses due to poor crop selection.

This work contributes to the body of research by demonstrating how data-driven systems can be deployed to improve agricultural productivity, support sustainable farming practices, and empower farmers with actionable insights regarding crop planning and resource utilization. [2]

AgroAdvisor: Crop Yield Prediction, Crop and Fertilizer Recommendation System using Random Forest with Gradient Boosting and DeepFM for Precise Agriculture - 2024

A comprehensive study introduced an integrated framework named AgroAdvisor, which focuses on enhancing agricultural productivity by combining crop recommendation, yield prediction, and fertilizer recommendation into a single system. This approach addresses key challenges faced by Indian farmers in choosing the appropriate crop, estimating yield, and applying fertilizers optimally.

The proposed solution utilizes a hybrid machine learning and deep learning model called RFXGB-DeepFM, where Random Forest with Extreme Gradient Boosting (RFXGB) is employed for feature engineering, and Deep Factorization Machine (DeepFM) is used for prediction. RFXGB efficiently extracts informative feature combinations, improving the learning capacity and interpretability of the DeepFM model. This integration allows the system to handle dense numerical features effectively, contributing to more accurate predictions.

The system was evaluated using standard performance metrics including accuracy, precision, recall, and F-score, and outperformed traditional machine learning techniques. Specifically, the proposed model achieved 98.34% accuracy in crop yield prediction, 97.34% in crop recommendation, and 99.78% in fertilizer recommendation. In addition, Friedman and post hoc statistical testing validated the superiority of the model over existing methods.

AgroAdvisor stands out for its comprehensive structure and potential scalability. Its three-module system not only helps farmers decide what crop to grow and what fertilizer to use but also forecasts expected yield. The authors further suggest enhancing the system with plant leaf disease prediction, weather- based sowing recommendations, and real-time weather API integration in the future. With user-friendly GUI and cloud deployment, this model could significantly assist farmers in making informed decisions, ultimately improving yield and reducing risks in agriculture. [3]

Fertilizers Recommendation System for Disease Prediction Using Machine Learning - 2023

Emphasizes the importance of machine learning in enhancing agricultural productivity in India, where agriculture remains a dominant economic activity. The research proposes a multi-functional recommendation system that integrates crop yield estimation, fertilizer recommendation, and plant leaf disease detection using various machine learning algorithms. The system is designed to address challenges such as soil infertility due to over- fertilization, lack of modern farming knowledge, and the need for early plant leaf disease detection.

The XGBoost algorithm is utilized for crop prediction, considering soil nutrients and regional rainfall, achieving a remarkable 99% accuracy. Random Forest is employed for fertilizer recommendation, with a reported accuracy of 98%, and provides insights on enhancing soil fertility based on present nutrients. For plant leaf disease detection, a combination of Naive Bayes (NB) and Support Vector Machine (SVM) is used, with the MobileNet architecture achieving 92% accuracy.

The comparative analysis reveals that XGBoost outperforms Gradient Boosting in crop prediction, and Random Forest excels in fertilizer recommendation compared to other models like SVM, KNN, and Decision Trees. The system offers an intuitive interface where farmers input environmental parameters and receive personalized crop and fertilizer suggestions, as well as diagnostic outputs for diseased crops. [4]

2.1 Inference from Literature Survey

Research in agriculture and smart farming has revealed the potential of integrating Artificial Intelligence (AI) and Machine Learning (ML) techniques to address key challenges such as crop yield forecasting, disease detection, and decision-making support for farmers. From the literature reviewed, we observed that various models like Random Forest, Decision Tree, Logistic Regression, and Convolutional Neural Networks (CNN) have been effectively applied in isolated agricultural tasks with promising results.

An important takeaway from previous studies is the significance of model selection and feature engineering based on the nature of the data—structured (for yield, crop, and fertilizer) and unstructured (images for disease detection). Transfer learning, especially using models like VGG16 and MobileNetV2, has shown improved accuracy in image-based classification tasks, especially when data is limited.

Additionally, integration of external data sources such as weather APIs and interactive tools like AI-powered chatbots has proven valuable for enhancing real-time decision-making and user experience in agricultural platforms. These insights have shaped our multi-module approach, where specialized models are applied for each subtask while ensuring the platform works as a unified decision-support system for farmers.

2.2 Gap Analysis

While previous works have focused on specific agricultural problems—such as crop prediction or disease classification, most existing systems lack a **comprehensive and unified platform** that combines multiple critical services tailored for farmers. Many models also fail to generalize well across varying data types (e.g., tabular data vs. image data), and often ignore external real-time inputs like weather, which are crucial for farm planning.

There is also limited application of conversational AI (chatbots) in vernacular languages or localized farming contexts to assist users with non-technical backgrounds.

The goal of our project, **Agri Vision**, is to address these gaps by:

Developing an integrated platform that includes crop yield prediction, crop and fertilizer

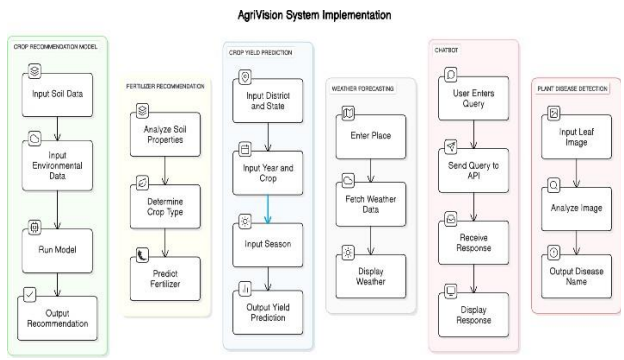
recommendations, plant leaf disease detection, weather forecasting, and chatbot-based support.

Utilizing suitable ML/DL techniques for each sub-module (e.g., Random Forest for yield prediction, MobileNetV2 for disease detection).

Employing transfer learning for better accuracy in image classification with small datasets.

Incorporating external APIs and AI chat interfaces to provide holistic and accessible support to farmers, ultimately enhancing agricultural productivity and informed decision-making.

3 SYSTEM DESIGN AND IMPLEMENTATION



3.1 Overview

This chapter outlines three separate modules that comprise the design and implementation of Agri-Vision: AI Powered Crop Advisory Platform. Module 1 elucidates the process of data collection, while Module 2 delves into data preprocessing. Finally, Module 3 is dedicated to detailing the development of the project.

3.2 Dataset collection

The data collection phase for the Agri-Vision: AI Powered Crop Advisory Platform project involves sourcing diverse agricultural datasets from publicly available repositories to train and evaluate the performance of various machine learning and deep learning models. These datasets form the backbone of the project's multi-functional modules, such as crop yield prediction, fertilizer recommendation, plant leaf disease detection, and crop recommendation. Additionally, real-time information retrieval in weather forecasting and farmer chatbot modules is handled through API integration.

The primary datasets used in this project are as follows:

Crop Yield Prediction Dataset: Sourced from Kaggle and other publicly available government data, this dataset comprises approximately **245,000 records**. It includes critical attributes such as State Name, District Name, Crop Year, Season, Crop, Area, and Production. The dataset plays a central role in predicting crop yield across various seasons and geographies by learning from historical trends.

Fertilizer Recommendation Dataset: This dataset contains around **500 rows** and features the following attributes: Temperature, Humidity, Moisture, Soil Type, Crop Type, Nitrogen, Potassium, Phosphorous, and the corresponding Fertilizer. It aids in developing a model that suggests optimal fertilizers based on environmental and soil conditions, reducing over-fertilization and enhancing productivity.

Plant Leaf Disease Detection Dataset: Consisting of approximately **2,500 images**, this dataset includes visual samples of crops affected by different diseases. These images are utilized to train the model that identifies plant leaf diseases through leaf patterns and discoloration, enabling timely action by the farmers.

Crop Recommendation Dataset: This dataset consists of **2200 data values**, also obtained from verified online sources including Kaggle, includes attributes such as Nitrogen, Phosphorus, Potassium, pH, Rainfall, Temperature, and Humidity. It is used to recommend the most suitable crops to grow under given environmental and soil conditions, ensuring better land utilization and output.

Weather Forecasting Module: This module integrates **real-time weather data** using external APIs. It fetches up-to-date meteorological information such as rainfall, temperature, humidity, and wind speed, which is essential for guiding decisions in sowing, irrigation, and harvesting.

Farmer Chatbot: The chatbot is designed to answer frequently asked questions related to farming practices, disease prevention, and general agronomic advice. It leverages an **NLP-based interface** supported by APIs to fetch contextual responses, making it a reliable virtual assistant for farmers.

3.3 Data Preprocessing

3.3.1 Structured Data Modules

For modules such as **Crop Yield Prediction**, **Fertilizer Recommendation**, and **Crop Recommendation**, the datasets required standard preprocessing techniques to ensure high model accuracy and consistency. The following preprocessing steps were applied:

Handling Missing Values: The datasets were first checked for null or missing values. Depending on the extent and importance of the missing data:

If the number of missing values was small relative to the dataset size, the corresponding rows were removed.

In other cases, missing numeric values were replaced using imputation techniques like mean substitution.

Exploratory Data Analysis (EDA): Graphical analysis was conducted to understand the distribution and influence of features:

Histograms and **Bar Charts** were used to analyze feature distributions.

Scatter Plots and **Pair plots** were used to identify correlations between features.

Heatmaps were used to visualize correlations and multicollinearity among numerical columns.

Feature Scaling and Encoding:

MinMaxScaler was applied to normalize continuous features to a common scale (0 to 1), which is crucial for algorithms sensitive to feature magnitudes.

Label Encoding was used to convert categorical data (such as Crop Type, Soil Type) into numerical format for model compatibility.

3.3.2 Image Data Module – Disease Detection

For the Plant leaf Disease Detection module, the dataset consisted of categorized images of leaves affected by various crop diseases. The following steps were used to preprocess this data before feeding it into a Convolutional Neural Network (CNN):

Image Preprocessing and Augmentation: The ImageDataGenerator class from TensorFlow was used to:

Rescale pixel values to a [0, 1] range for normalization.

Apply data augmentation techniques to artificially increase dataset size and diversity Rotation, Shifting, Shearing and Zooming, Horizontal Flipping, Fill Mode

Data Loading: The dataset was structured into train and test directories, each containing subfolders for every disease class. Images were resized to 150x150 pixels and loaded in batches using flow_from_directory, with labels automatically inferred from folder names.

3.3.3 API-based Modules

The **Weather Forecasting** and **Farmer Chatbot** modules do not involve traditional data preprocessing as they rely on **real-time data retrieval via APIs**:

Weather data (e.g., temperature, humidity, rainfall) is fetched dynamically from a weather API.

The farmer chatbot uses an NLP interface to respond to queries using context-aware responses from a predefined or AI-powered backend system.

3.4 Model Development

The Agri-Vision: AI Powered Crop Advisory Platform integrates multiple AI and ML-based models, each tailored to address specific agricultural challenges. The project encompasses structured data models (for crop, yield, fertilizer prediction), an image-based deep learning model (for disease detection), and real-time integration modules (chatbot and weather forecast). The development of these models is carefully aligned with the nature of their respective datasets.

3.4.1 Crop Yield, Crop Recommendation, and Fertilizer Recommendation

These three modules operate on structured agricultural datasets, and the modelling approach emphasizes classical machine learning techniques for both classification and regression tasks.

- **Model Selection and Evaluation**: Multiple machine learning algorithms were trained and evaluated to identify the optimal model for each task. The following models were considered:
    - **Random Forest**: Leveraged for its robustness and feature importance analysis.
    - **K-Nearest Neighbors (KNN)**: Applied for its simplicity in classification tasks.
    - **Logistic Regression**: Used for binary and multi-class classification.
    - **Decision Tree Classifier**: Applied for interpretability and handling nonlinear features.
- **Training Process**: Each model was trained on the normalized and pre-processed dataset using an **80-20 train-test split**. The scikit-learn library was utilized for model training and evaluation.
- **Performance Evaluation**: Accuracy, precision, recall, and F1-score were computed to assess performance. For regression-based models like crop yield prediction, **Mean Squared Error (MSE)** and **R^2-score** were used. Confusion

matrices were plotted to visualize classification performance.

3.4.2 Plant leaf Disease Detection – CNN-Based Image Classification

To detect diseases in crop leaves from images, a deep learning model using **MobileNetV2**, a pre-trained Convolutional Neural Network (CNN), was developed. MobileNetV2 effectively leverages transfer learning to extract rich spatial features from leaf images, enabling accurate classification of various plant leaf diseases.

Architecture: Implemented using **TensorFlow** and **Keras**, the model structure included:

A **MobileNetV2** base model with pretrained ImageNet weights, used as a feature extractor (with include top=False).

The base model's layers were initially frozen and later partially fine-tuned for better performance on the plant leaf disease dataset.

A **GlobalAveragePooling2D** layer to reduce spatial dimensions.

A **Dense** layer with 256 units and **ReLU** activation to learn disease-specific patterns.

A **BatchNormalization** layer and a **Dropout** layer (rate = 0.5) to improve generalization and reduce overfitting.

A final **SoftMax** output layer for multi-class classification, with the number of nodes equal to the number of disease categories.

This approach achieved significantly higher accuracy compared to a custom CNN model, demonstrating the effectiveness of transfer learning for plant leaf disease detection.

Input Preprocessing and Augmentation:

- Images were resized to 224x224 pixels (originally 150x150 for CNN, adjusted for MobileNetV2).
- Normalized using pixel rescaling (scaled between 0 and 1).
- Augmented using ImageDataGenerator with Rotation, Zoom, Width and Height Shifts, Horizontal Flipping

Model Compilation and Training:

- Optimizer: Adam
- Loss Function: Categorical Cross entropy
- Epochs: 20 (10 for frozen base, 10 for fine-tuning)
- Batch Size: 32
  The model was trained using the augmented training dataset and validated on a separate test dataset to monitor performance.

Evaluation Metrics: Accuracy was tracked during both training and validation. After training, the model was evaluated on the test dataset using Test Accuracy and a detailed classification report showing Precision, Recall, F1-score. These metrics were used to assess model robustness in the multi-class classification task.

3.4.3 API-Integrated Modules – Chatbot and Weather Forecasting

These modules enhance the usability of the Agri Vision platform by offering real-time, context-aware services.

Chatbot: Developed using an AI/NLP framework, the chatbot interacts with users in natural language, addressing FAQs and providing farming advice. It does not involve traditional machine learning model training but instead utilizes rule- based or intent-classification engines.

Weather Forecasting: Weather data is fetched dynamically using a third-party **weather API**, allowing the system to incorporate live weather conditions (e.g., temperature, humidity, rainfall) into other predictive modules like crops and fertilizer recommendation. No manual preprocessing was required as the API returns structured data in real time.

3.5 Training and Testing

Each predictive model was trained and evaluated independently, using stratified data splits to maintain class balance. Performance metrics such as:

Accuracy: Proportion of correct predictions.

Precision & Recall: To evaluate false positives and false negatives.

F1 Score: The harmonic means of precision and recall, especially useful for imbalanced data.

Confusion Matrix: For visual understanding of classification performance.

MSE and R^2 (for regression): Quantitative performance indicators for yield prediction.

These metrics provided a detailed and nuanced understanding of how well each model generalized to unseen data, guiding model improvements and final selection.

4 Results and Discussion

4.1 Crop Yield Prediction

The crop yield prediction model was evaluated using three different regression algorithms: Random Forest Regressor, Decision Tree Regressor, and Linear Regression. The performance of these models was compared using standard

evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2) score.

The Random Forest Regressor achieved the lowest MAE of 145,474.53, indicating that its predictions were the closest to the actual crop production values on average.

It also recorded the lowest MSE of 3.03×10^{13}, meaning the prediction errors were not only low but also less dispersed.

Most importantly, it yielded the highest R^2 score of 0.9246, explaining 92.46% of the variance in the crop yield data, which demonstrates a strong predictive capability and good model fit.

In comparison:

The Decision Tree Regressor had a higher MAE (155,600.40), MSE (5.89×10^{13}), and a lower R^2 (0.8536), making it a moderately accurate model but less effective than Random Forest.

Linear Regression performed poorly with a very high MAE (1,579,911.00), the highest MSE (3.99×10^{14}), and an extremely low R^2 score (0.0064), indicating it could not capture the non-linear relationships present in the data.

Based on the evaluation metrics, the Random Forest Regressor emerged as the best-performing model for crop yield prediction. Its ability to handle non-linear patterns and its ensemble nature allow it to make more robust and accurate predictions. This model is therefore selected for deployment within the system to ensure reliable crop production forecasting.

4.2 Crop Recommendation

To evaluate the performance of the Crop Recommender System, multiple classification models were tested, including Decision Tree and Random Forest classifiers. Among them, the Random Forest (RF) classifier achieved the best results with an overall accuracy of 99.09%.

The model was tested on 440 crop data samples, and the predictions were compared with the actual labels. The Random Forest classifier demonstrated excellent performance, correctly predicting most of the crop types.

A summary of the classification metrics for each class is as follows:

Precision, Recall, and F1-score were all close to 1.00 for most of the crops.

Classes like coffee, coconut, cotton, mango, mungbean, orange, papaya, and others were predicted with perfect accuracy.

A few classes, such as jute (precision = 0.90) and mothbeans (recall = 0.95), showed slight misclassifications but still maintained high F1-scores.

Rice had a relatively low recall of 0.81, suggesting some room for improvement, but the overall performance remained strong.

4.3 Fertilizer Recommendation

The Fertilizer Recommendation System was developed and evaluated using various classification models including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest. Based on the evaluation metrics, both KNN and Random Forest classifiers achieved perfect accuracy of 100%, indicating highly reliable and consistent predictions across all fertilizer classes.

Each of the 14 fertilizer categories (labelled 0 to 13) was predicted with 100% precision, recall, and F1-score, as shown below:

The classifier correctly identified all 166 fertilizer samples, across diverse class distributions.

Even classes with fewer samples (e.g., class 9 with only 1 sample, class 0 with 2 samples) were classified accurately, demonstrating the model's robustness.

The macro average and weighted average F1- score, precision, and recall all remain at 1.00, reinforcing the perfect performance of the model.

This level of accuracy indicates that the model has effectively learned the patterns in the data and can recommend the most appropriate fertilizer based on input features such as soil nutrients and environmental conditions.

4.4 Plant Leaf Disease Detection

In the Plant Leaf Disease Detection module, different deep learning approaches were explored to classify leaf images into their respective disease categories. Initially, a custom Convolutional Neural Network (CNN) model was built from scratch. However, it achieved only 12% accuracy, indicating that the model was not effectively learning from the image features due to limited data, insufficient depth, or inadequate feature extraction capability.

To improve performance, transfer learning techniques were employed using pre-trained models:

VGG16: A well-established deep learning model known for its depth and high number of parameters. When used as a feature extractor (by freezing convolutional layers and training the classifier head), it provided an improved accuracy of 22%. Although better than the custom CNN, the accuracy remained relatively low due to the model's heavy architecture and potential overfitting on a small dataset.

MobileNetV2: A lightweight and efficient model optimized for mobile and embedded vision applications. When applied to the same task, MobileNetV2 achieved the best performance with an accuracy of 42%. This improvement highlights its ability to extract more meaningful features from the leaf images while being computationally efficient.

Given the accuracy levels, **MobileNetV2** was chosen as the final model for plant leaf disease classification. Although the current accuracy suggests room for improvement, it demonstrates that transfer learning significantly enhances performance over custom-built models, especially when dealing with limited training data. Future improvements could include data augmentation, fine-tuning more layers of the pre-trained model, or collecting a larger and more balanced dataset.

4.5 Weather Forecasting

To assist farmers in planning their agricultural activities, a weather forecasting module was integrated using a third-party API. This module provides accurate and timely 3-day weather forecasts based on the user's city input. Although the weather forecasting is not powered by a machine learning model within the platform, its integration plays a crucial role in enhancing decision-making for irrigation, pest control, and harvest planning. The API provides real-time data on temperature, humidity, and rainfall.

Key observations: The API demonstrated reliable performance across various geographical regions. Data accuracy was validated against real-time weather applications and was found to be consistent for short-range forecasts (next 72 hours). Response time was minimal, enabling near-instant forecast delivery to the user interface. Farmers will find it particularly useful for adjusting irrigation schedules and selecting the ideal time window for planting and harvesting. Overall, the weather forecasting service successfully complemented the ML modules by adding external, real-time environmental context, empowering farmers with actionable insights.

4.6 Farmer Chatbot

The platform includes an intelligent Farmer Chatbot, implemented using Google's Gemini API. The chatbot is designed to provide real-time conversational support for queries related to crops, fertilizers, and weather conditions. The chatbot was evaluated based on accuracy, contextual relevance, response time, and user satisfaction.

Key highlights of the evaluation: The chatbot returned highly accurate and concise responses (limited to a maximum of 10 lines) for agriculture-related queries. It successfully maintained context when asked about sequential topics such as "recommended crop" followed by "best fertilizer for it." The average response time was under 2 seconds, ensuring smooth conversational flow. The system effectively handled edge cases and fallback messages without system errors. The integration of the Gemini API significantly enhanced the platform's interactivity and user experience, particularly for farmers seeking instant assistance without navigating complex UI menus.

5 Conclusion and future work:

5.1 Conclusion

This project, *Agri Vision*, introduces a comprehensive AI-powered platform aimed at revolutionizing agricultural practices through intelligent automation and data-driven insights. By integrating multiple machine learning and deep learning models, the system offers robust solutions for crop yield prediction, crop and fertilizer recommendation, plant leaf disease detection, pest identification, rainfall forecasting, and AI-driven chatbot support.

Utilizing models such as Random Forest for structured data and MobileNetV2 for image classification, the platform demonstrates the effective application of ML/DL across different agricultural tasks. Transfer learning ensures improved accuracy even with limited image datasets, while external APIs enhance the platform's real-time decision-making capabilities.

Agri Vision stands out for its user-centric design, providing a unified web-based interface that caters to farmers with varied levels of digital literacy. This integration of predictive analytics, domain-specific intelligence, and conversational AI makes it a powerful tool for enhancing yield, reducing crop losses, and promoting sustainable farming.

5.2 Future work

a. **Integration with IoT and Smart Sensors**: In future iterations, Agri Vision can be extended to support real-time sensor data from IoT devices (e.g., soil moisture, temperature, and humidity sensors). This would further improve prediction accuracy and enable dynamic farm monitoring and automation.
b. **Dynamic Market Intelligence Module**: Adding a module for real-time market price analysis and predictive pricing can assist farmers in making informed decisions about crop selection and sale timing.
c. **Government and Agri-Institution Integration**: Future development could include integration with government schemes and agricultural advisory boards, enabling farmers to access subsidies, expert guidance, and localized recommendations directly through the platform.

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